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Consulting the community on the future of HPC in Canada

MAY 2010

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Compute Canada's Mission

1. To help ensure that Canadian researchers have easy and efficient access to HPC technologies, applications, and skills on a national level; and
2. To represent and promote Canadian academic high performance computing nationally and internationally to governments, funding agencies, and society in general.

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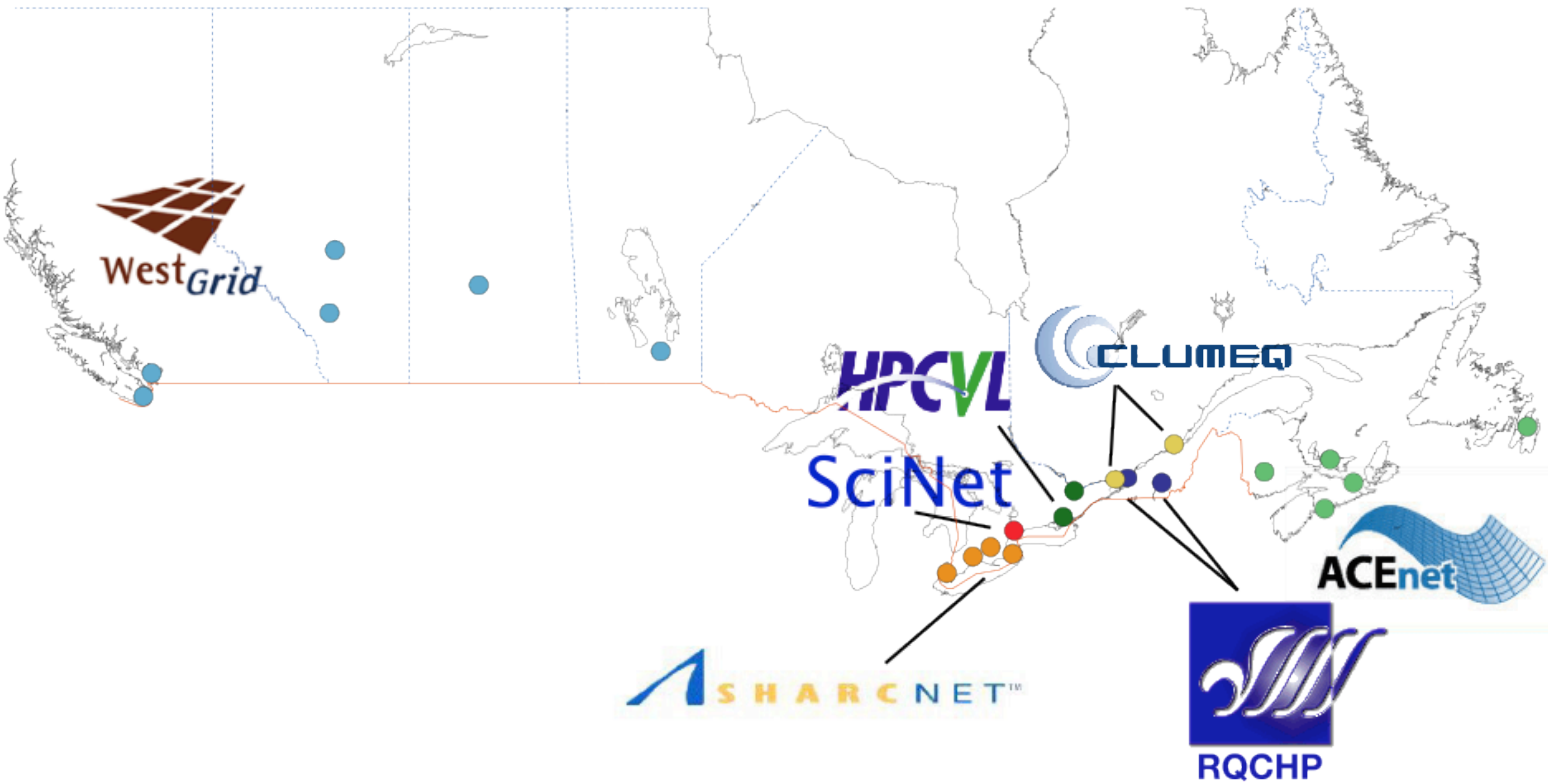
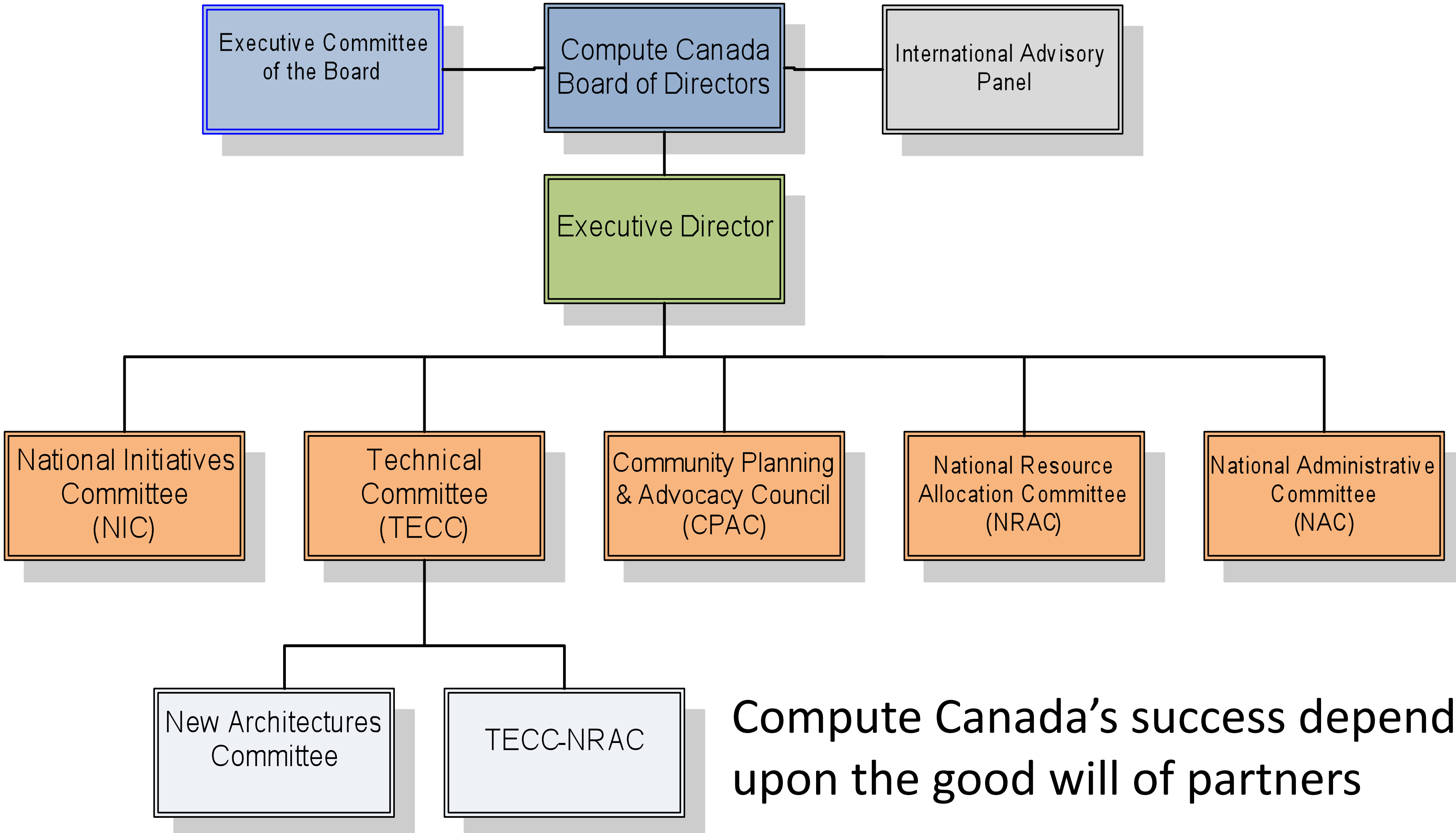
The seven regional consortia



Organization Chart



Compute Canada's success depends upon the good will of partners

Update

- Equipment RFPs, purchases & installation – January 2008 – present
- Compute Canada Database goes online – June 2009
- NRAC Call for Proposals – July 2009
- Mid-Term Review Meeting with International Review Panel – May 25, 2010
- NRAC 2nd Call for Proposals – September 2010
- NPF-2 Call expected December 2010

Business as usual – Not

- The federal deficit
- No “entitlement”
- Funding will be a competitive process
  – no “set-aside” for Compute Canada
- Focus on federal S&T Strategy priority areas:
  – Environmental science and technologies
  – Natural resources and energy
  – Health and related life sciences and technologies
  – Information and communications technologies

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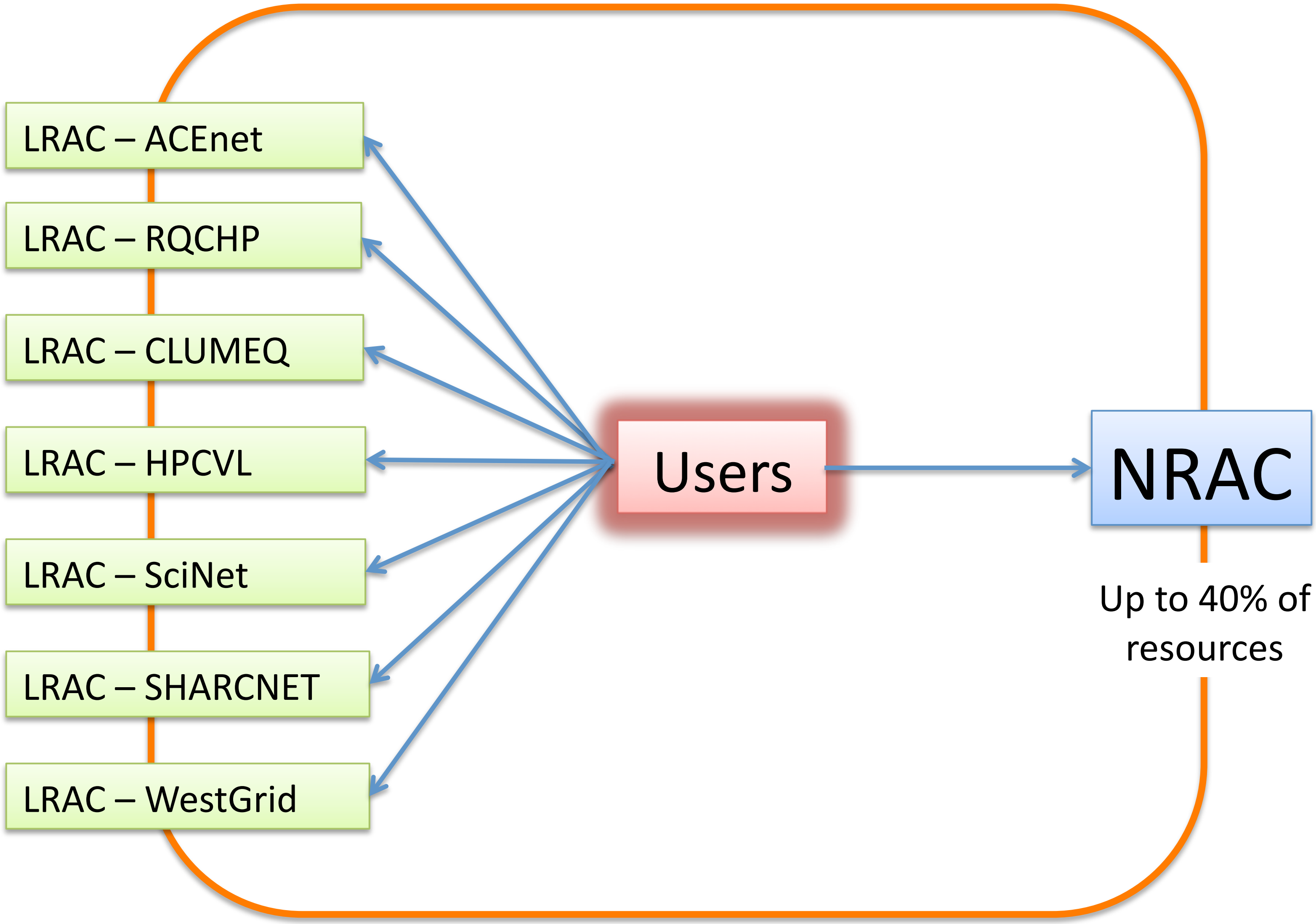
Enabling Canadian Research

- Focus is on ensuring the ability of the Canadian researcher to do better research by:
  - Using the right facility (wherever in the country)
  - Using the right methods
  - Using the right amount of compute cycles
  - Having the HQP support they need
  - Knowing HPC capacity will grow in lock step with their research
- HOW ARE WE DOING?

Theme 1: Resource Allocation

approx. 20 minutes

Resource Allocation Committees



Allocation process

- Criteria for NRAC projects
  - A project that spans more than a single consortium as a result of collaboration among researchers from different consortia, where there is a need to share resources among them and the level of resources required by each researcher is significant;
  - A project that requires a substantial fraction of the resources of a consortium or exceeds the limits set for a category of access by the consortium;
  - A project that requires resources in several consortia because it depends upon a distributed test bed even if a large amount of resources is not required.

- Allocations periods for all RACs are now synchronized to calendar years

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First NRAC allocations : results

System	consortium	% NRAC	% LRAC
NPF systems			
Colosse	CLUMEQ	23.0%	–
Checkers	WestGrid	17.0%	61.0%
Bugaboo	WestGrid	27.0%	70.0%
Orcinus	WestGrid	8.0%	78.0%
Victoria	WestGrid	35.0%	–
GPC-GigE	SciNet	34.0%	29.0%
GPC-IB	SciNet	2.0%	72.0%
TCS	SciNet	22.0%	40.0%
Non NPF systems			
Hound	SHARCNET	–	2.0%
Saw	SHARCNET	0.3%	
HPCVL	HPCVL	3.0%	

Questions

1. Is the current resource allocation process working for you?
2. Do you have any questions about the NRAC or LRAC processes?

Theme 2: User support

approx. 20 minutes

Types of support

- Basic Access help
  - accounts, connections, libraries, job submission, etc.
- Porting code to a specific architecture
- Optimization and parallelization
- Training sessions
- Writing documentation
- Help with visualization or collaboration tools
- Other?

Staff levels

	Management	Infrastructure support	User support
ACEnet	3.00	5.25	5.00
CLUMEQ	0.80	3.00	4.00
HPCVL	3.50	6.00	4.00
RQCHP	1.80	5.25	6.78
SciNet	4.00	6.00	6.00
SHARCNET	5.97	8.25	11.30
Westgrid	5.20	13.80	10.55
Total	**24.27**	**47.55**	**47.63**
% of total	20.3%	39.8%	39.9%
% of budget	24.9%	34.4%	40.7%

Questions

1. Are there circumstances under which researchers would be willing to pay for services (extensive technical consultation/unique software/other)?
2. Are you receiving the technical support you require to optimize your research?
3. Should, and if so, how should Compute Canada address the security and confidentiality issues of various research groups?

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Theme 3: Computing resources

approx. 20 minutes

National Platform 1

- $60M CFI + $60M Provinces + in-kind
- 23 compute systems + storage systems
- 22 locations
- Deployment between Spring 2009 and 2011

- Fund distribution across consortia was based roughly (and solely) on historical funding (or lack thereof)

Computing equipment

Name	site	type	cores	storage (TB)	accessibility
Colosse	Laval	capability	7 680	500	Jan. 2010
–	Queen's	storage	–	650	Dec. 2009
Bugaboo	SFU	capacity	1 280	1 200	May 2009
Orcinus	UBC	capacity	3 072	–	Jun. 2009
–	Lethbridge	fat node cluster	384	40	Summer 2010
Checkers	U. of Alberta	capacity	1 280	100	Jun. 2009
–	U. of Calgary	capability	4 096	140	Summer 2010
TCS	U. of Toronto	capability	3 328	–	May 2009
GPC	U. of Toronto	hybrid cluster	30 240	–	Aug. 2009
–	U. of Toronto	storage	–	1 500	
Silo	U of Saskatchewan	storage	–	600	Mar. 2009
Nestor	Victoria	capability	2 304	–	Jun. 2010
Hermes	Victoria	capacity	672	–	Jun. 2010
Pleiades	Victoria	storage	–	900	Jun. 2010
Coming equipment					
	U. of Manitoba	capability			Fall 2010
	U. of Alberta	SMP + new arch.			Late 2010
	Concordia	capacity			Fall 2010
	U. de Sherbrooke	capability			Fall 2010
	McGill	capability			Late 2010
	U. de Montréal	capability + GPU			Early 2011

Constraints on future equipment distribution

- Provinces/regional agencies contribute to 50% of cash amount
  - Single Tier 1 facility impossible in this context
- Many institutions contribute to construction and/or operations, in particular power costs
- Many large data centers in Canadian Universities (much has been spent on renovations already)
- The quality of the science is not questioned by CFI: all is about *efficiency* and *organization*

Questions

1. Compute Canada must reduce the number of data centres. What mechanisms/criteria should be used to determine the location of future (NPF-2) HPC equipment? Should external reviews be conducted? In what way, positively or negatively, would such a reduction impact your research?
2. Could Compute Canada and its partner consortia be organized in a way that would better address the needs of researchers to access machines across the country in an efficient and timely manner? Do researchers need to be geographically close to machines? Or is the critical factor having technical support geographically close?

Questions

3. How important is it that Compute Canada provide the full capability HPC ecosystem to the research community (even if you might not use it yourself)?
   - Should there be a Tier 1 component in the next NPF?
   - Should Compute Canada buy computing from the cloud?
   - What specialized services, software, infrastructure or resources should be available?

Theme 4: Organization

approx. 20 minutes

Compute Canada and consortia

- Compute Canada must be more than the sum of its parts
- Currently : staff of one (ED)
  Depends on consortia for volunteer work on most issues
- TECC : collaboration of technical staff from all consortia
- Insufficient funds for effective coordination

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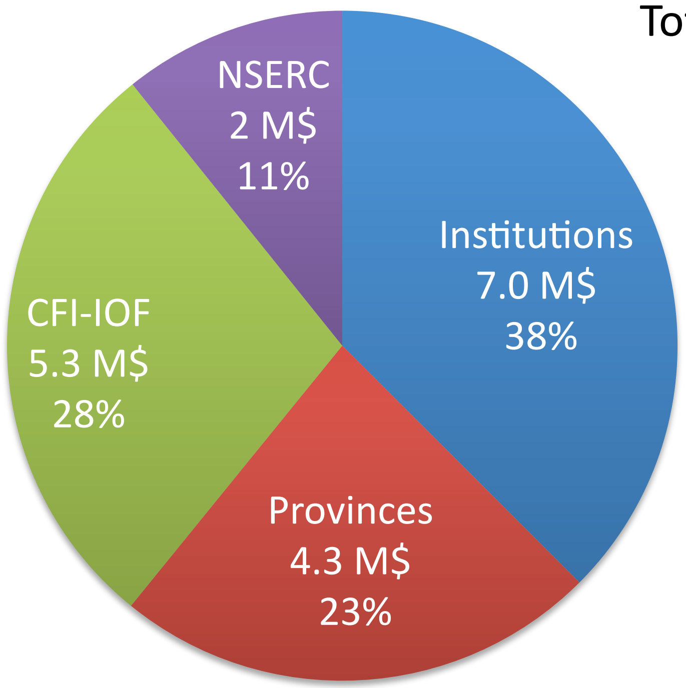
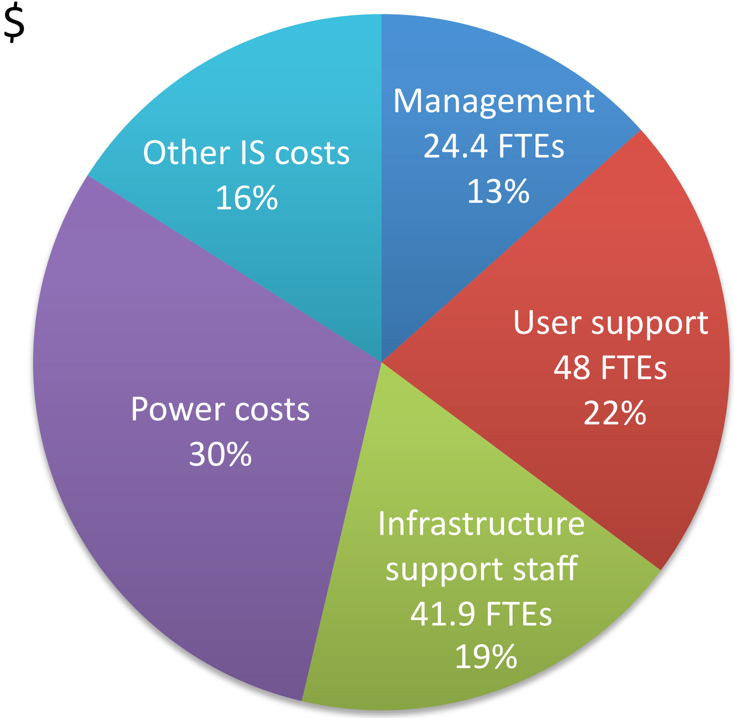
Cost of Operations

Revenues



Total : 18.6 M$

Expenses



Based mostly on actual revenues and expenses in 2009 + some future projections on power costs. Not a sustainable steady state situation.

Questions

1. What changes to the structure of Compute Canada are necessary to ensure that decisions and outcomes are in the best interests of researchers across the country? In particular, how should the roles of regional consortia evolve in relation with Compute Canada?

2. There is a single consortium for the western provinces and a single consortium for the Atlantic provinces. What would you see as the benefits or potential drawbacks of a single consortium for each of Ontario and Quebec? What organizational model would ensure the best support for researchers?

3. Would HPC in Canada benefit if Compute Canada represented all HPC with members from government departments and the private sector? Would the "academic" voice be stronger with additional partners?

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Your questions here

Backup material

Resource use by discipline (2009)